

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 9-10

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*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Key Formulas This Week

- $R_a^2 = 1 - \frac{\frac{SSE}{n-(p+1)}}{\frac{SST}{n-1}} = 1 - (1 - R^2) \left(\frac{n-1}{n-(p+1)} \right)$
- Mallow's $C_p = \frac{1}{n} (SSE_p + 2ps_{full}^2)$
- $BIC = \frac{1}{n} (SSE_p + \ln(n) ps_{full}^2)$
- Out-of-Sample: $SSPE = \sum_{test} (y_i - \hat{y}_i)^2$
- $CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i$
- LOOCV: $PRESS = \sum_{i=1}^n (y_i - \hat{y}_{(i)})^2 = \sum_{i=1}^n \left(\frac{e_i}{1-h_{ii}} \right)^2$

Iterative Modeling

- The iterative approach to model building consists of **model formulation** based on preliminary data analysis, **fitting** parameters, then **diagnostic checking** to reveal if any assumptions are violated, and then revising/iterating these steps to create the model.
- With a final model, **inferences** can be made.

Auto Variable Selection

- As the # of predictors grows, the # of potential models grows towards infinity quickly (2^p).
- Solution: Automated procedures can give a few interesting models to focus on.
- **Stepwise regression** is a process that uses t -tests to check significance of explanatory variables to ____

Stepwise Regression

- **Forward stepwise selection** adds the next best predictor for the overall model (already including the first stage variable), repeating until all statistically significant variables at a chosen significance level α have been added.

Stepwise Regression 2

- **Backward stepwise selection** includes all variables (full model) initially, then _____

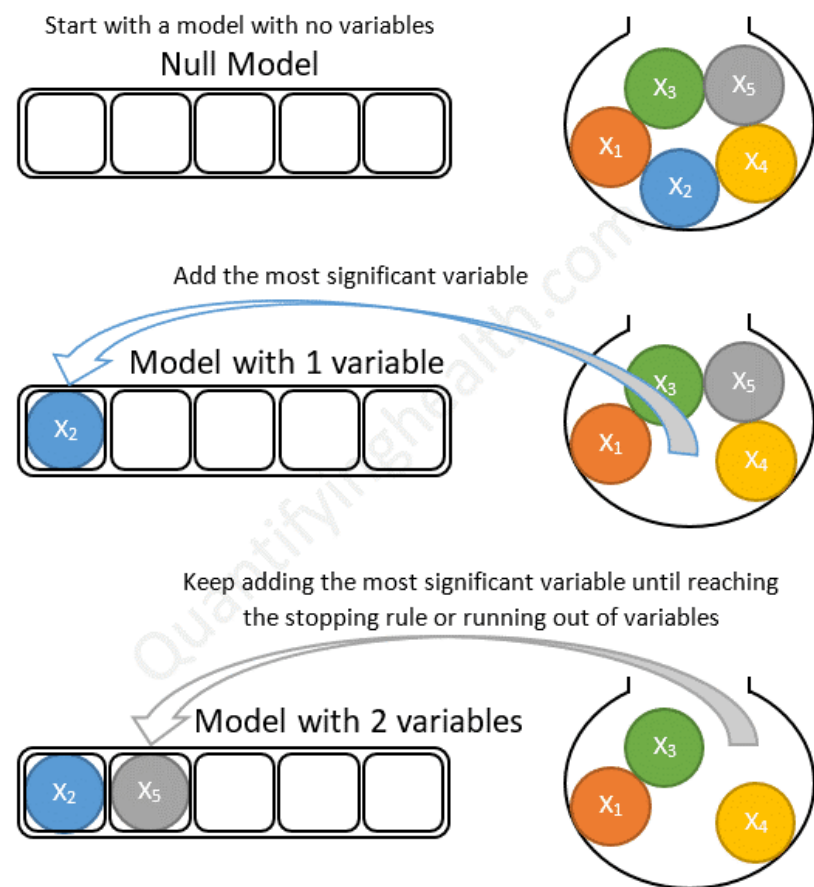
Stepwise Regression 3

- Both forward stepwise and backward stepwise selection can be done without a stopping rule, meaning the forward model starts with 0 parameters and ends with all p , and the backward model starts with all p and ends with 0.
- Forward and Backward SS lead to
- Then, some goodness of fit statistic (OOS prediction error, R_a^2 , C_p , AIC , BIC) determines the best model.

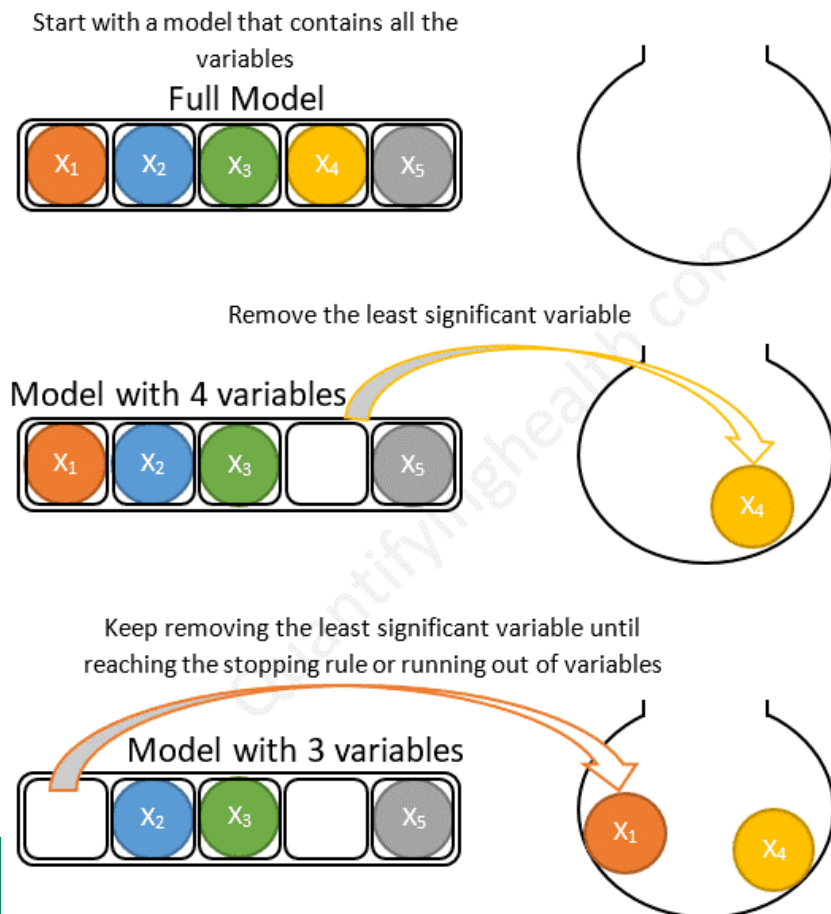
Stepwise Visual

• <https://quantifyinghealth.com/stepwise->

Forward stepwise selection example with 5 variables:



Backward stepwise selection example with 5 variables:



Mixed Stepwise

- In a **mixed stepwise regression** algorithm,
 - 1. Consider all regressions that add the best explanatory variable for the model (that exceeds a set significance threshold).
 - 2. Delete the 1 worst explanatory variable for the model, if it is below the set significance threshold.
 - 3. Repeat until the model stabilizes.

Best Subset Selection

- The **best subset selection** algorithm searches
-

Automated Disadvantages

- All automated methods (forward stepwise, backward stepwise, stepwise regression, best subset) suffer from these issues:
- Susceptible to _____
- Chosen model is not _____
- Does not account for any _____

Stepwise Regression 4

- ***Advantage:***
- Speed and efficiency:
- ***Disadvantages of Forward & Backward:***
- Does not search all _____
- Uses only one criterion, the _____
- Sequence of significance tests can make the _____

Forward Stepwise

- ***Advantage:***
- Forward stepwise selection can be used in high-dimensional cases when _____
- ***Specific Disadvantages:***
- Once a variable enters the model, _____
- Can miss the _____

Backward Stepwise

- ***Advantage:***
 - Better able to capture joint variable effects, since _____
- ***Specific Disadvantage:***
 - Cannot be used in _____

Best Subset Selection 2

- ***Advantage:***
 - Searches all 2^p possible combinations, minimizes training SSE
- ***Specific Disadvantage:***
 - Can take considerable _____

Best Subset Selection: Example with 3 Variables



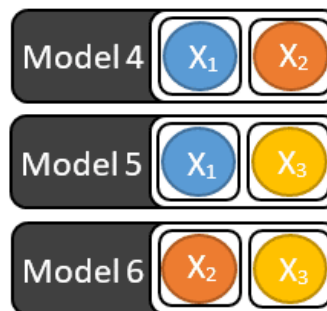
Step 1: Consider All Possible Models

By listing all possible combination of variables

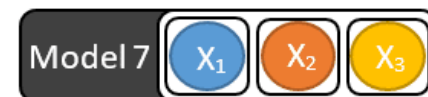
Models with 1 variable:



Models with 2 variables:



Models with 3 variables:



Step 2: Identify the Best Model of Each Size

By choosing the one with the lowest sum of squared errors or the highest R^2

Best model with 1 variable:



Best model with 2 variables:



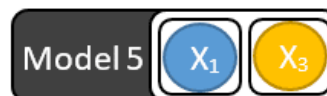
Best model with 3 variables:



Step 3: Identify the Best Overall Model

By choosing the one with the lowest AIC (or BIC) or the highest adjusted R^2

Best overall model:



- Null model not shown.
- <https://quantifyinghealth.com/best-subset-selection/>

Concept Check #1

- I. The predictors in the k -variable model identified by forward stepwise are a subset of the predictors in the $(k + 1)$ -variable model identified by forward stepwise.
- II. The predictors in the k -variable model identified by backward stepwise are a subset of the predictors in the $(k + 1)$ -variable model identified by forward stepwise.
- III. Best subset selection results in nested models.
- IV. Forward and backward stepwise selection can result in the same best model when # of predictors p equals number of observations n .
- V. Forward stepwise selection may miss joint effects of variables compared to backwards stepwise.
- VI. Unlike best subset selection, backwards stepwise cannot account for business insights.
- VII. Forward stepwise selection cannot be used in high-dimensional cases.

1. Stepwise Illustration

- Several linear regression models are fitted to a dataset with 30 observations. Given the results, identify the variables used in the final model under forward stepwise selection, at the 5% significance level. How many models need to be fit under forward stepwise vs. best subset?

Explanatory Variables	Residual Sum of Squares
None	16
x_1	13
x_2	12
x_3	14
x_1, x_2	10
x_1, x_3	9
x_2, x_3	11
x_1, x_2, x_3	8

1. Stepwise Illustration 2

- $F = \frac{\frac{(SSE_0 - SSE)}{q}}{\frac{SSE}{n - (p + 1)}} = \frac{\frac{RSS_{reduced} - RSS_{full}}{df_{diff}}}{\frac{RSS_{full}}{df_{full}}}$
- $t_{n-(p+1)} = \sqrt{F}$ with p based on the H_a model in 1 parameter difference cases
- FSS fits $1 + 0.5p(p + 1)$ models. $BestSS$ fits 2^p .

Adjusted R^2

- Goodness-of-fit statistics measure how well the fitted model aligns with the actual data.
- R_a^2 is one such metric but can only be used for linear models. Higher is better.

$$\bullet R_a^2 = 1 - \frac{s^2}{s_y^2} = 1 - \frac{\frac{SSE}{n-(p+1)}}{\frac{SST}{n-1}} =$$

- n : # of data points
- p : # of explanatory predictors

Example 1

- Linear regression models are fitted to a dataset with 30 observations. Identify the model candidates in forward stepwise selection vs. backward stepwise selection.
- Which model is chosen under best subset selection using adjusted R^2 as the criteria?

Explanatory Variables	Un-Adjusted R^2
None	0%
x_1	15%
x_2	20%
x_3	25%
x_1, x_2	45%
x_1, x_3	30%
x_2, x_3	35%
x_1, x_2, x_3	50%

Example 1 Sol

- Adjusted $R^2 = 1 - (1 - R^2) \left(\frac{n-1}{n-(p+1)} \right)$
- At the same # of predictors, R^2 gives the same rankings as adjusted R^2 .

Example 1 Sol 2

- Best chooses the $\{x_1, x_2, x_3\}$ model

Explanatory Variables	Adjusted R^2
None	0
x_1	0.12
x_2	0.17
x_3	
x_1, x_2	
x_1, x_3	0.25
x_2, x_3	0.3
x_1, x_2, x_3	

Goodness of Fit

- The most common statistical criteria of model quality for comparison are:
- Mallow's C_p , which is equivalent to the Akaike information criterion (AIC) in linear regression,
- Bayesian information criterion (BIC)
- Models with _____

Mallow's C_p / AIC, BIC

- AIC / Mallow's C_p =
 - s_{full}^2 is MSE from a model with all explanatory predictors
 - SSE_p : Error sum of squares for p predictors
- BIC =

Mallow's C_p / AIC, BIC 2

- Mallow's C_p is linear model specific but AIC/BIC apply to any _____
- BIC penalizes more for $n \geq 8$ and in this case will never _____

Concept Check #2

- Determine if each statement is True.
- I. AIC can provide a direct estimate of the test error.
- II. Given models with the same p , AIC and BIC will always select the same model.
- III. For $n \geq 8$, BIC will select a model with smaller p than AIC.
- IV. Mallow's C_p is an unbiased estimator of the test MSE.
- V. Small BIC values generally indicate a model with low test error.

2. GOF Illustration

- You fit the following models to a dataset of 25 observations and 3 explanatory variables to predict # of TV shows that will be cancelled.
- Determine the best model under adjusted R^2 , Mallows's C_p , and BIC .

Model	# of Variables	Error SS
I	None	100
II	x_1	72
III	x_1, x_2	65
IV	x_1, x_2, x_3	60

2. GOF Illustration 2

- Mallow's $C_p = \frac{1}{n} (SSE_p + 2ps_{full}^2)$
- $BIC = \frac{1}{n} (SSE_p + \ln(n) ps_{full}^2)$
- $R_a^2 = 1 - \frac{\frac{SSE}{n-(p+1)}}{\frac{SST}{n-1}} = 1 - \frac{s^2}{s_y^2}$

2. GOF Illustration 3

- $R_{a1}^2 = 1 - \frac{\frac{100}{25-(0+1)}}{\frac{100}{25-1}} = 0$ as expected

- $R_{a2}^2 =$

- $R_{a3}^2 = 1 - \frac{\frac{65}{25-(2+1)}}{\frac{100}{25-1}} = 0.291$

- $R_{a4}^2 = 1 - \frac{\frac{60}{25-(3+1)}}{\frac{100}{25-1}} = 0.314$

2. GOF Illustration 4

- $s_{full}^2 = MSE_{full} =$
- Ignoring the $\frac{1}{n}$
- $C_{p1} = 100 + 2(0)(2.8571) = 100$
- $C_{p2} =$
- $C_{p3} = 65 + 2(2)(2.8571) = 76.4$
- $C_{p4} = 60 + 2(3)(2.8571) = 77.1$

2. GOF Illustration 5

- $BICnum_1 = 100 + \ln(25) (0)(2.8571) = 100$
- $BICnum_2 =$
- $BICnum_3 = 65 + \ln(25) (2)(2.8571) = 83.4$
- $BICnum_4 = 60 + \ln(25) (3)(2.8571) = 87.6$

Example 2 (CAS)

- Under best subset vs. forward stepwise regression vs. backward stepwise regression with AIC used to compare models with different # of parameters, determine the candidate models and chosen models under each approach.

Model	# Predictors	Parameters	R^2	AIC
1	0	I (intercept)	0	2.0
2	1	I, 1	0.49	1.6
3	1	I, 2	0.5	1.5
4	1	I, 3	0.48	1.7
5	2	I, 1, 2	0.58	1.2
6	2	I, 1, 3	0.6	1.0
7	2	I, 2, 3	0.56	1.3
8	3	I, 1, 2, 3	0.7	1.1

Example 2 Sol

Example 3

- You fit several models to 100 observations.
- Age, Income, Points are continuous variables.
- Degree is a categorical variable w/8 levels.
- Parent is a categorical variable w/2 levels.
- $\sum_{i=1}^{100} (y_i - \bar{y})^2 = 130^2$
- Determine which model Adjusted R^2 chooses vs. Mallows's C_p vs. BIC.

Model	Predictors	R^2
I	Age, Income	0.70
II	Age, Income, Degree	0.79
III	Age, Income, Points, Parent	0.74
IV	Age, Income, Points, Degree, Parent	0.80

Example 3 Sol

- $R_a^2 = 1 - \frac{s^2}{s_y^2} = 1 - (1 - R^2) \left(\frac{n-1}{n-(p+1)} \right)$
- Mallow's $C_p = \frac{SSE_p + 2ps_{full}^2}{n}$
- $BIC = \frac{1}{n} (SSE_p + \ln(n) ps_{full}^2)$
- $MSE = s^2 = \frac{SSE}{n-(p+1)}$
- $R^2 = 1 - \frac{SSE}{SST}$

Model Validation

- **Model validation** seeks to find the most appropriate model and address data snooping.
- The best technique is **out-of-sample validation**, meaning _____
- Testing set = Validation set = Hold-out set
- Using the test dataset, calculate sum of squared prediction errors: $SSPE = \sum_{test} (y_i - \hat{y}_i)^2$



Model Validation 2

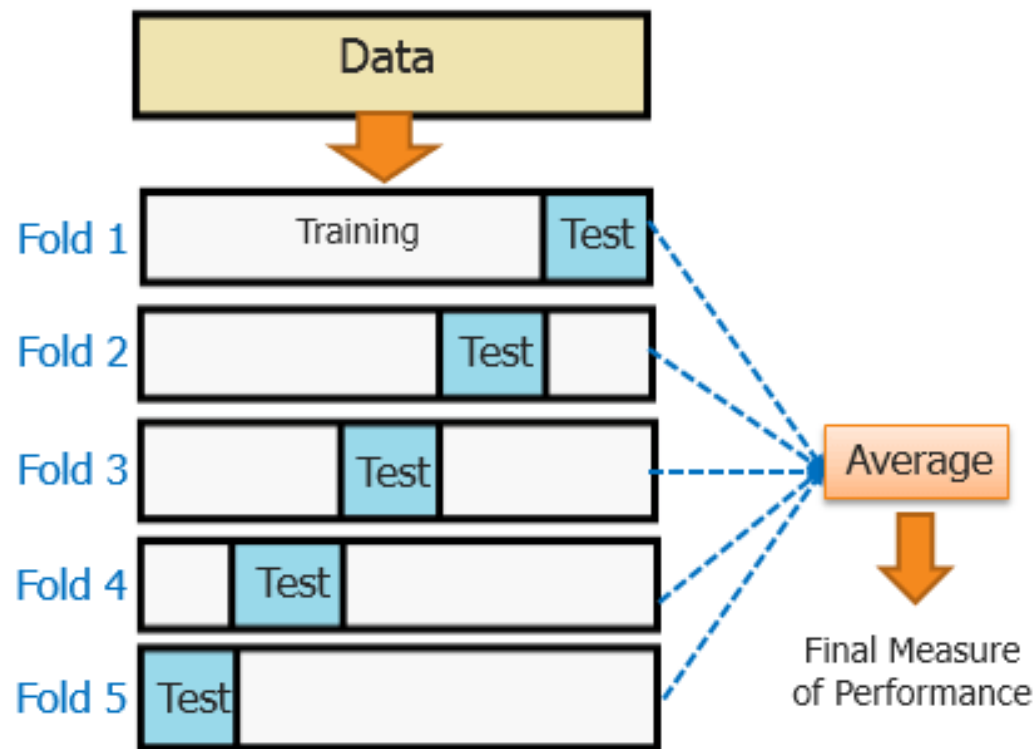
- **Disadvantages** of out-of-sample validation:
- Time consuming to test *SSPE* for many models (true of all validation)
- Model depends on random sampling and will vary across analyses.
- Choice of training vs. test sample size is not clear.
- Use of smaller training set leads to _____

Cross-Validation

- k -fold cross-validation is similar in that the data is split into a predetermined k groups, and one group is _____
- $CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i$
- The model with the lowest $CV_{(k)}$ is best.

Cross-Validation 2

- Each observation is validated exactly once and trained on $k - 1$ times.



Cross-Validation 3

- Advantage: More stable and less biased estimate of test MSE because $CV_{(k)}$ is an _____
- Disadvantage vs. true out-of-sample validation: There's not true independence between the estimation and prediction aspects

LOOCV

- Leave one-out cross-validation (LOOCV) is an automated technique where the test sample is one observation and the training sample is the remainder.
- Predicted Residual Sum of Squares: $PRESS = \sum_{i=1}^n (y_i - \hat{y}_{(i)})^2$
 - $\hat{y}_{(i)}$: Prediction of i^{th} observation using all points except i^{th} observation in the model
- Under MLR, $PRESS =$
- $CV_{(n)} = \frac{1}{n} PRESS$ is the estimated test mean square error
- The model with the lowest $PRESS$ or $CV_{(n)}$ is best.

LOOCV 2

- Advantages:
- Less bias than out-of-sample validation due to using
- One test error rate for each model (deterministic) that does not depend on random splitting as out-of-sample validation or k-fold do
- Disadvantage:
- Computationally more intense

Bias-Variance Tradeoff

- Bias:
- Variance:
- Tendency to overestimate test error rate:
- Commonly, 5-fold or 10-fold cross-validation is used as a balanced trade-off.
- LOOCV is a special case of k -fold cross validation with

2 Competing Goals

- The goal of model selection is always to balance:
- Including more explanatory variables to not _____
- Including fewer explanatory variables to keep the model _____
- If the true relationship is roughly linear and n is large relative to p , linear regression performs well with low bias.

Modeling Process Links

- 1. Clean and process data pre-modeling.
- 2. Automated variable selection using the different methods and metrics can provide a few excellent starting candidate models.
- 3. Modelers iteratively try out more models with interactions, transformations, etc.
- 4. Model validation vets the candidate models to find which generalizes best. Model diagnostics show potential assumption violations or issues.
- 5. Use judgment to select the final model and estimate its parameters using the full dataset before entering production.

Concept Check #3

- Determine if each statement is True.
- I. k -fold validation requires fitting k separate models.
- II. LOOCV requires fitting only 1 model if least squares regression is used.
- III. LOOCV requires fitting $n - 1$ models if least squares regression is not used.
- IV. The validation set error is robust and most likely will not change when its calculation procedure is repeated.
- V. The validation set error tends to underestimate the test error.
- VI. The validation set approach tends to produce bias.
- VII. The validation set approach tends to produce high variance as an estimator of the test error.

3. 1SE Illustration

- Find the optimal # of predictors using the one-standard-error rule.

# Predictors	CV Error	Standard Error
1	10	2.0
2	8.0	1.6
3	7.0	1.8
4	6.0	1.7
5	6.2	1.9
6	6.5	1.5

3. 1SE Illustration 2

- The one-standard-error rule is set at the minimum cross-validation error plus one standard error.

Example 4

- The model $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ is fit to the data:

y_i	1	2	3	4	5
x_{1i}	0.5	1	0.75	0.25	0.5
x_{2i}	4	6	2	3	1

- $(X^T X)^{-1} X^T y = \begin{bmatrix} 5.02 \\ -0.22 \\ -0.59 \end{bmatrix}$

- $X(X^T X)^{-1} X^T = \begin{bmatrix} 0.35 & 0.23 & -0.03 & 0.42 & 0.03 \\ 0.23 & 0.87 & 0.16 & -0.1 & -0.16 \\ -0.03 & 0.16 & 0.55 & -0.13 & 0.45 \\ 0.42 & -0.1 & -0.13 & 0.68 & 0.13 \\ 0.03 & -0.16 & 0.45 & 0.13 & 0.55 \end{bmatrix}$

- Calculate the LOOCV estimate for the test MSE.

Example 4 Sol

- $CV_{LOOCV} = \frac{1}{n} \sum_{i=1}^n \left(\frac{e_i}{1-h_{ii}} \right)^2$

HW Note

- We won't cover regularization this semester, but you may see question parts refer to lasso.
- Variable selection effectively sets excluded predictor coefficients to 0.
- Ordinary least squares rarely gives exact 0 estimates, but methods like lasso can.

Summary Check

- How Forward Stepwise vs. Backward Stepwise vs. Best Subset Selection work and pros/cons
- Formulas for Adjusted R^2 , AIC/Mallow's C_p , BIC
- How validation set vs. k -fold cross-validation vs. LOOCV work, pros/cons, bias-variance tradeoff